

The Ephemeris

The Official Publication of the San Jose Astronomical Association

December 2024
Volume 35 Issue 4

C/2023 A3 Tsuchinshan-Atlas

Page 15

70th Anniversary Potluck

Page 8

Swap Meet Recap

Page 13

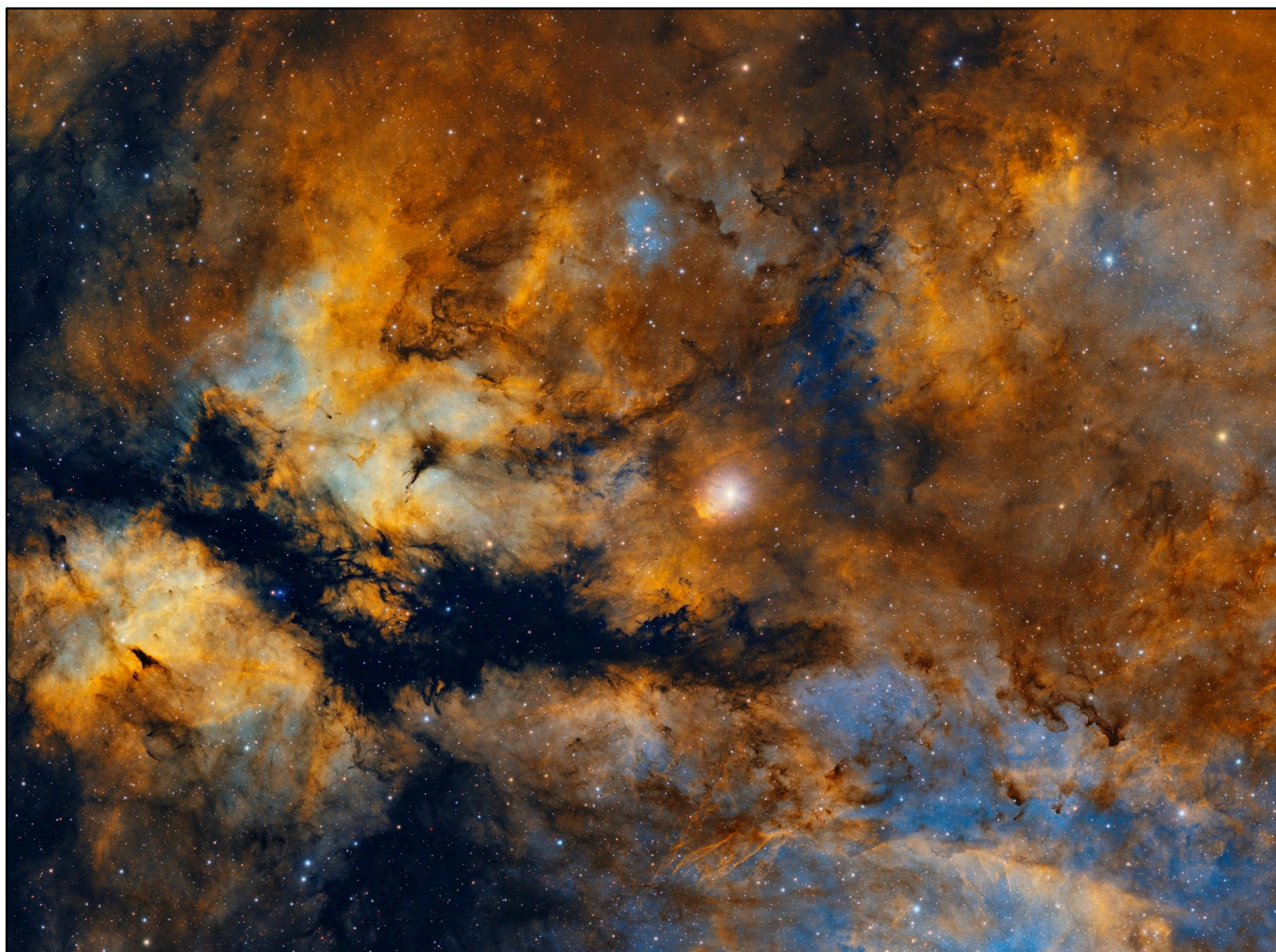




Left: Lunar Occultation of Saturn by Bryan Murahashi. Taken on September 17th, 2024 from San Jose (Bortle 7 skies).

A nearly full moon passed in front of Saturn early in the morning, in an event known as an "occultation" - when one object temporarily covers up another one from our viewing perspective. Shown here is Saturn just before the moon passed in front of it at 4:07am. Saturn's rings are clearly visible, along with the dramatic shadows on the lunar edge.

Taken with Nikon Z8 camera, Tamron 150-500mm F15-6.7 Di III VC VXD lens at 1/30s, f/8, 500mm, ISO 100.



Top: Sadr Region in Cygnus Mosaic by Alex Ranous. Over 33 hours of integration time taken from Redwood City with AP Starfire GTX 130, AP 1100-GTO-AE, QHY 268M, Astrodon filters. Acquired with NINA and processed with BlurX, StarX, NoiseX, PixInsight, and APCC. <https://www.astrobin.com/6mjepi/>

Front Cover: Comet C/2023 A3 Tsuchinshan-ATLAS by Francesco Meschia. About 44 minutes of integration time taken from Russian Ridge OSP with AT65EDQ, ASI 294MM Pro, AP Mach 1 GTO, and Astronomik RGB filters. More details on Astrobin: <https://www.astrobin.com/10j4s1/>

Back Cover: The Andromeda Galaxy by Carl Svensson. Over 10 hours of integration time in ZWO branded RGB filters, taken with an SV80ST scope, ASI 1600MM Pro camera, and EQ6-R Pro mount from a Bortle 3 location in the Sierra Foothills. <https://www.astrobin.com/fjy96/B/>

Inside This Issue

Member Programs	4
Public Programs	4
Announcements	5
From The SJAA President	6
Board Minutes	6
Join Hubble's Night Sky Challenge	7
SJAA 70th Anniversary Potluck	8
Kids Corner	10
Observing Highlights	10
SJAA Recent Events	11
Ready Now Results from Our Satisfaction Surveys!	12
Swap Meet Recap	13
School Star Party Award	13
Spot the King of Planets!	14
C/2023 A3 Tsuchinshan-ATLAS	15

Corrections

The following corrections are made to the Sept. 2024 Ephemeris

- On page 2, the image labeled M31 is actually M51.
- Larry Zurbrick's name is misspelled in several places

Calendar

BOARD MEETINGS + SPEAKER Saturday 12/14, 01/11, 02/15 Hogue Park, 6:00 - 7:30pm <i>Guest speaker follows meetings</i>	STARRY NIGHTS STAR PARTY Saturday 01/25, 02/22 RCDO, 6:00pm
SOLAR SUNDAY Sunday 12/01, 01/26, 02/23 Hogue Park, 2:00 - 4:00pm	IMAGING SIG Tues 12/17, 01/21, 02/18 Online, 7:30pm
IN-TOWN STAR PARTY Friday 12/20, 01/24, 02/21 Hogue Park, 7:15pm	FIX-IT SESSIONS Sunday 12/01, 01/26, 02/23 Hogue Park, 4:00pm

Please refer to the SJ Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events) where you can find specific event times, locations, maps and possible cancellation due to weather.

SJAA Contacts

President	Larry Zurbrick
Vice President	—
Treasurer	Rob Jaworski
Secretary	Swami Nigam
Director 1	Vini Carter
Director 2	Radhika Gupta
Director 3	Ken Miura
Director 4	Wolf Witt
Director 5	Swami Nigam
Director 6	Rob Jaworski
Director 7	Peter Melhus
Director 8	Glenn Newell
Director 9	Marianne Damon
Ephemeris Editor	Carl Svensson
Binocular Stargazing	Ed Wong
Fix-it Program	Larry Zurbrick
Hands on Imaging	Glenn Newell
Imaging SIG	Hy Murveit
Astronomy 101	Carl Svensson
In-Town Star Party	Muditha Kanchana
Library	David Kershaw
Loaner Program	Muditha Kanchana
Memberships	Kashaul Gangakhedkar
Publicity	Beth Johnson
Quick Start	Jonathon Lawton
Donations	Rich Gregor
Solar	Wolf Witt
Starry Nights	Joe Fragola
School Events	Jon Anderson
Refreshments	Marianne Damon
Social Manager	Beth Johnson
Speakers	—
Webmaster	Satish Vellanki
Volunteer Coordinator	Nancy Lieu

 @sj_astronomy
 @sjastronomy


SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly. Articles and Observation Reports for publication should be submitted by e-mail to ephemeris@sjaa.net. Please only submit original content (articles or photos) of your own authorship. Copyrighted content from other authors, including images, will be rejected.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Member Programs

SJAA LIBRARY

SJAA offers another wonderful resource: a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments to librarian@sjaa.net.

TELESCOPE FIX-IT SESSION

Fix-It Day, sometimes called the Telescope Tune Up or the Telescope Fix-It program, is a really simple service the SJAA offers to members of the SJAA community for free, though it's priceless. Headed up by Larry Zurbrick, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

QUICK START PROGRAM

The Quick START Program, headed up by Jonathan Lawton, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick START sessions are generally held quarterly. The Quick START program has resumed under its new lead, Jonathon Lawton.

<http://www.sjaa.net/programs/quick-start/>

LOANER PROGRAM

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

ASTRO IMAGING SIG

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month over Zoom, often on the 3rd Tuesday evening of the month. We usually have a guest speaker presenting on some topic related to astro-imaging. Talks begin at 7:30pm. There is an opportunity to socialize and discuss imaging starting at 7pm when we start the Zoom call. Meeting topics, dates and times are posted on the SJAA Astro Imaging mail list in Google Groups. Past meetings are available on [YouTube](#).

<http://www.sjaa.net/programs/imaging-sig/>

ASTRO IMAGING WORKSHOPS AND FIELD CLINICS

Not to be confused with the Imaging SIG group, this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Hogue Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

Public Programs

ASTRONOMY 101

Astronomy 101 covers various topics related to beginner astronomy and star gazing. The popular Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Hogue Park. This is a regular, monthly session, run by Carl Svensson. Content varies from "what's up in the night sky" to basics of star-gazing and telescopes to specific topics of interest. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

SJAA By The Numbers

Membership	393	(+0)
X Followers	945	(+8)
Facebook Followers	829	(+14)
Instagram Followers	162	(+23)

SCHOOL STAR PARTY

The San Jose Astronomical Association conducts evening observing sessions (commonly called “star parties”) for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill. Contact SJAA’s Program Coordinator for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SOLAR OBSERVING

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time and location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

IN-TOWN STAR PARTY (ITSP)

This popular public star party is held every month, at our headquarters in Houge Park in San Jose. Enjoy celestial wonders from our very own backyard. Our volunteers are more than happy to show you the grate views through their telescopes. Check our Meetup page for specific dates and times. All are welcome!

Announcements

SJAA APPAREL STORE is now open! Get your official, branded apparel at <https://sjaa.printify.me>. Shirts, hats, hoodies, tote bags, and even phone cases are available for your shopping delight. Ships quickly and makes for a great holiday gift!

PRESIDENT LARRY ZURBRICK assumed his new role after former president Aroon Nair stepped down. We thank Aroon for all of his hard work, and welcome Larry to this new position!

SJAA ON KKUP On September 18th, our very own Marianne Damon gave an interview to Ms V. of She-Radio on KKUP. Thank you, Marianne for representing SJAA on the air waves!



San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243
(Must be 18 years or older to apply)

New Membership or Renewal?

(check one)

☐

NEW

☐

RENEWAL

Membership Type

(check one)

☐

\$20 Membership with online Ephemeris

☐

\$30 Membership with printed Ephemeris

(mailed to address below)

NAME

ADDRESS

CITY/STATE/ZIP

PHONE

E-MAIL

Bring this form to any SJAA meeting or mail it to: SJAA, P.O. Box 28243 San Jose, CA 95159-8243

This newsletter is available online at <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net. Join or renew online at <http://www.sjaa.net/join-the-sjaa/>

From The SJAA President

The View from Below the Skies



Welcome to a new addition to the SJAA Ephemeris: comments from your SJAA President on the various aspects of the SJAA and amateur astronomy.

I'm always impressed by the depth of knowledge our members openly share with the public on a broad range of subject matter. Everything from detailed equipment operation to the latest breaking news in astrophysics and cosmology. We do this at a large number of public outreach events in the field and at our Houge Park meeting site. The dedication our program leaders have to this mission is astounding in terms of their preparatory work and time spent with the public. Our volunteers who participate in public star parties with their telescopes or binoculars and sharing their knowledge of the sky also need to be commended on their dedication. The patience, conscientious consideration, and carefully considered answers they provide answering the public's

endless questions is an inspiration.

This leads me to another point. Many of our Program Coordinators do not have an heir apparent training in the background in case of a Program Coordinator having a scheduling conflict or stepping back from the leadership position. Is there a particular SJAA program that you regularly participate in as a volunteer? Have you considered taking on more of a leadership role? Have you considered that a particular program could be suspended or end in the event the Program Coordinator is no longer able to continue? Do note that before stepping up to take on a role as Assistant Program Coordinator be willing to commit for your time and talent for the longer term. The Program Coordinator will be investing their time and talent in you to hand you the reins of running the program. Give it some thought and then step up.

As always, ad astra.

Board Minutes

SEPTEMBER 14, 2024

UPDATED MEMBERSHIP FORMS Mike Jaworski has provided a prototype implementation of the membership form using Gravity forms plugin and walked us (Swami, Satish, Rob) through its implementation. Swami did quite a lot of testing. However, on trying to format the emails from this form to look like our original Welcome emails, we ran into the same issue as with the current form - the emails failed to send. After tweaking the contents of the Welcome emails, Swami was able to get the emails working again - both with the original form as well as with Gravity forms. So, the custom form is working again for now. We should still switch to Gravity forms, after some further testing and connecting to production instance of PayPal.

OCTOBER 19, 2024

SANTA CRUZ MUSEUM OF NATURAL HISTORY

First Friday was a successful event. As per the museum, about 240 people attended. Game wheel was once again a big hit. A gentleman named Miguel spent a lot of time at our table - he was pleasant company. He also introduced us to Rick Nolthenius who teaches at Cabrillo College and would be a good person for our speaker event. This led to an invitation to join the Santa Cruz STEAM Expo in March 2025.

COLLABORATION WITH ASTRO REVOLUTION

We had a follow up call with the Astro Revolution's president and vice president and brainstormed on

possible avenues for collaboration, now that they have had a chance to attend some of our events, and some of us (Wolf, Swami) have attended their workshops. Over the next few weeks, we will meet again and put together a proposal for collaboration, which we will then share with the board. The President and Vice-President of Astro Revolution have been invited to meet with the board at a board meeting - tentatively they are planning to attend the Dec 14 meeting.

NOVEMBER 16, 2024

MORGAN HILL LIBRARY Requestor Brandy M. Morgan Hill Oakwood school was selected to make contact with the ISS as part of community engagement; they have asked the library to host space/STEM programs to get the community excited about the contact. Report from Jon: It was a lot of fun, very different from SSP, with music playing, wine & cheese. The organizers loved the event, and would love to do it again. However, the location was not so good, with lots of trees around. If we organize with this group again, we will need to find another location.

MENDOZA RANCH CHAIR We have one member, Doug Loyer, who expressed interest in being the Mendoza Ranch permit contact person. Larry plans to discuss with the board whether to move forward with Doug. Joe Fragola also has received input from several members during the recent Starry Nights that expressed willingness to assist with Mendoza Ranch activities. Larry will continue working with Doug and others to finalize the MR chair.

Excerpts from this quarter's board meetings are above. Full minutes are available online at <https://www.sjaa.net/about/board-meeting-minutes/>



Join Hubble's Night Sky Challenge

Image Credit: Carl Svensson

In celebration of the Hubble Space Telescope's 35th anniversary, NASA invites all amateur astronomers and night sky enthusiasts to participate in Hubble's Night Sky Challenge. This exciting year-long adventure offers a unique opportunity to explore the same celestial objects that Hubble has captured from space, bridging the gap between amateur and professional astronomy in a fun, interactive way.

Every month in 2025, participants will receive a curated list of targets to observe through their telescopes. These targets include a diverse array of celestial wonders, from distant galaxies to stunning nebulae, and even the planets within our own solar system. The challenge provides a structured way to engage with the night sky, offering both seasoned stargazers and beginners the chance to hone their observation skills and deepen their understanding of the cosmos.

What makes this challenge unique is the ability to compare your observations with the breathtaking images captured by the Hubble Space Telescope. Hubble, orbiting high above Earth's atmosphere, has provided us with some of the most detailed and awe-inspiring views of the universe. By participating in this challenge, you can experience the thrill of discovering these wonders yourself. Keep in mind that ground-based observations made with our amateur telescope equipment will not be as bright or colorful as Hubble's images. Likewise, visual observations will not be as bright as digital images, but of course you get the

bragging rights that come with having your eyes glimpse the actual photons from these amazing objects. Give yourself the best chance of success by finding a nice, dark spot to observe. Be sure to take note of the apparent size and magnitude of each object, and make sure you choose appropriate equipment for each target.

To add a touch of friendly competition, NASA will recognize successful observers each month in the form of official certificates. These certificates come in two flavors: "silver" and "gold." Silver certificates can be obtained by making a minimum number of observations each month. The requirements also list an outreach event, but it is not clear if that is an alternative requirement, or if it is in addition to a report of the observations. Gold certificates and a special pin are awarded to those who meet requirements throughout the year. Submissions should be made directly to the NASA contact, Richard Benson, at rlbenson6691@comcast.net. See the table below and the Astro League webpage for requirements: <https://shorturl.at/hspep>. Stay tuned to the official NASA webpage for updates on the targets throughout the year: <https://shorturl.at/kZzX5>.

It's also worth noting that there are other challenges run by NASA and the Astronomical League. For example, the "Parker Solar Probe 2.0 Special Observing Award" is currently up for grabs. Complete observations of the sun for Perihelion #22 on December 24th. Successful observations could net you a certificate and a nifty pin.

SILVER REQUIREMENTS (MONTHLY)

- Observe at least one object from the list in the proper month
- Sketch or image the object, and identify it
- Perform one or more outreach events

GOLD REQUIREMENTS

- Observe four or more objects from each month
- Sketch or image the object, and identify it
- Perform four or more outreach events in 2025

Send Submissions To

Richard Benson, 3105 Lykes Dr. NE, Albuquerque, NM 87110
Telephone: 505-340-6691 Email: rlbenson6691@comcast.net

SJAA 70th Anniversary Potluck

This year, SJAA celebrates its 70th anniversary! To mark the occasion, we held a potluck dinner at our home base, Houge Park. Marianne Damon and her Snacks team set up a neatly streamlined buffet table, along with a communal, U-shaped seating area for all members and guests.

Attendees were treated to a variety of delicious food. Main entrees included barbecue ribs, pasta with home grown tomatoes, and cauliflower humus, along with great appetizers like the savory lentil cake. And the desserts - oh, the desserts! We had donuts, cookies, and the main attraction: a great, big cake emblazoned with the SJAA 70th anniversary logo.

Even better than the food, of course, was the company. It was a real pleasure to see current and former active SJAA members, including our two guests of honor: Robert Garfinkle and Robert Fingerhut. Prior to the start of the potluck, Bob Garfinkle presented the SJAA Board with all three volumes of his latest book, "Luna Cognita: A Comprehensive Observer's Handbook of the Known Moon." This, along with some of his other books, are available in the SJAA library today!

Both Roberts addressed the potluck attendees and shared history and fun stories from the club's past. Robert Fingerhut shared this particularly interesting story about Dr. Gregory's C8 telescope:

He (Dr. Gregory, former SJAA president and honoree of

the Gregory Award) had one of the first Celestron 8 telescopes in Northern California. In fact, it was serial number 5. It arrived at his house, and there was a big palm print on the corrector plate. So, he called up Tom Johnson, who founded Celestron, and he came up and cleaned the telescope. Then we found out why (there was a palm print). The telescope had been first sent to Playboy magazine for a photoshoot before it was sent to Dr. Gregory's house. Someone there at the photoshoot had put their palm on it.

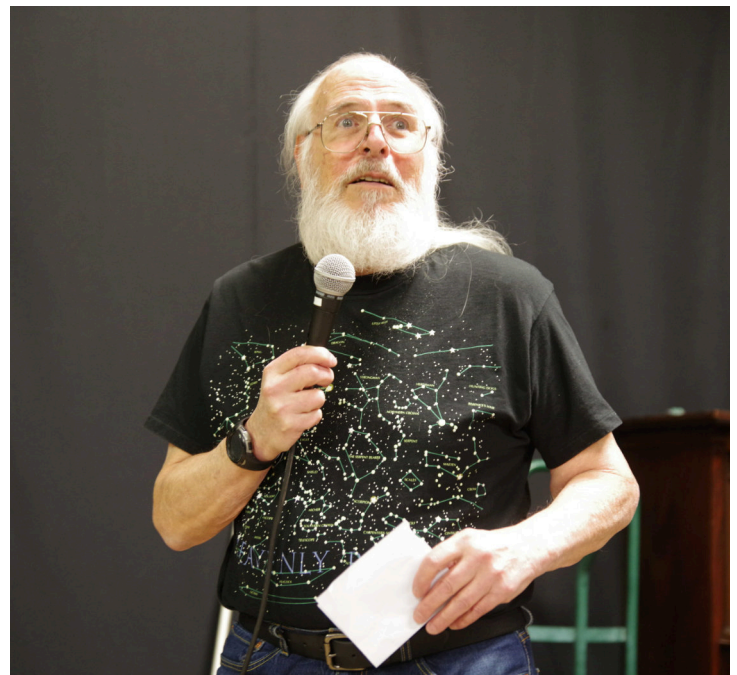
Along with the great food and people, we also had a great trivia game, complete with prizes. Marianne compiled a list of questions which shed light on our club's distinguished membership and accomplishments. Here's a small sample of fascinating trivia:

- Several members have astronomical objects named after them, including Don Machholz (12 comets and one asteroid), Dr. Peter Jenniskens (asteroid), and Dr. Elinor Gates (asteroid).
- Don Machholz invented the Messier Marathon, and Robert Garfinkle was the first to publish it in a book.
- In 1999, SJAA made a telescope at Astronomy Days at The Tech Museum in just 14 hours!
- SJAA sponsored the very first Cal Star Party (CSP) at Lake San Antonio in 2000.

Special thanks go to Marianne and everyone who made this truly special event such a great success!



Bob Fingerhut addressing the SJAA membership. Image credit: Ken Miura.



Bob Garfinkle addressing the SJAA membership. Image credit: Ken Miura.



Setting up for the potluck. Turns out, SJAA members are really efficient at setting out tables and chairs!



Marianne set up a prize table with our latest SJAA apparel. Players spun the wheel and answered trivia for prizes.



Bob Fingerhut (left), SJAA President Larry Zurbrick, and Bob Garfinkle cut the cake.



Food time!



There was no shortage of scrumptious desserts, including a specially made 70th anniversary cake.

Observing Highlights

12/07

Jupiter at Opposition

At its closest approach to Earth, this is the best time to view and image Jupiter.

12/13

Geminids Meteor Shower

The king of meteor showers lasts from the 7th to 17th, but contends with a full moon.

12/21

December Solstice

At 09:17 UTC the Sun will be directly over the Tropic of Capricorn, 23.44° south.

12/21

Ursids Meteor Shower

Producing 5-10 meteors per hour, this minor shower contends with the moon.

12/25

Mercury at Greatest Elongation

At 22° from the Sun, this is the best time to view Mercury just before sunrise.

01/03

Quadrantids Meteor Shower

Best of the showers this quarter, it peaks overnight on the 3rd.

01/10

Venus at Greatest Elongation

At its furthest distance of 47.2° from the Sun, this is the best time to view Venus.

01/16

Mars at Opposition

This closest approach to the Earth will be the best time to view Mars.

New Moon: 12/01, 12/30, 01/29, 02/28

Full Moon: 12/15, 01/13, 02/12

Source: <http://www.seasky.org/astronomy/astronomy-calendar-2024.html>

KIDS CORNER

Word Search

- ▶ Tsuchinshan
- ▶ Kuiper Belt
- ▶ Snowball
- ▶ Tail
- ▶ comet
- ▶ ATLAS
- ▶ Coma
- ▶ Oort Cloud

Q U X J F M H H E A W I Y C M
P U Q V B C F A E N D F Y G R
S Q E K W A O F T F S U K N T
X W O Y Q C O M C L B I H G S
T W O C G G M F E F A J V A F
S U R T P N I I C T G S Y A R
U O T A Q S N O W B A L L S U
C I C I H W J H R W U W R H J
H K L L S C L T C H T W Q L E
I H O O I M O D S F N O V T E
N L U J F B X M C T N L X O K
S V D L Q A J K A A Z B T W X
H Z S J M M B D Z J B P Q O J
A X L K U I P E R B E L T T E
N T S H J W J H Z P W Z V H Z

Fun Facts

- ▶ Comets are cosmic snowballs of frozen gases, rock, and dust that orbit the Sun.
- ▶ When frozen, they are the size of a small town.
- ▶ When a comet gets close to the Sun, it heats up and spews dust and gases into a giant glowing head larger than most planets.
- ▶ The dust and gases form a tail that stretches away from the Sun for millions of miles!

Source:
<https://science.nasa.gov/>

SJAA Recent Events

We hold all kinds of cool events throughout the year. Here's just a sample of what we've been up to since the last issue! -Carl

Solar Observing

We held our solar observing session for November on the 10th. It was a great day in the park -- clear skies, active Sun, not too hot -- with a good volunteer crew on SJAA club equipment plus a few club members with their own gear. Thanks to everyone who came out to share with our visitors! On Sunday, the Sun showed a few more spots than it did a day later (see right). Prominences were interesting although not nearly as large as the filaments. Wolf has been gathering and testing the equipment for our new solar telescope station, an 80mm refractor with a sunspot projection funnel. Thanks to Tom and Rich for providing the mount and telescope from the donations inventory, while this particular projection funnel is Wolf's 3D-printed creation. *(Credit: Wolf)*

Telescope Workshop

On Sunday, November 17th we held our first ever, "All About Telescopes Workshop." Over the course of three, hour-long segments, Wolf, Swami, and Carl S. presented material describing: how different kinds of telescopes work, how lenses and mirrors produce an image, how mounts work, the characteristics of eyepieces, how binoculars work, and most importantly, how to evaluate all of this equipment in order to meet your observing needs. In between sessions, we provided a selection of cookies while we hosted brief show-and-tell sessions with all of the telescopes and binoculars on display.

Starry Nights

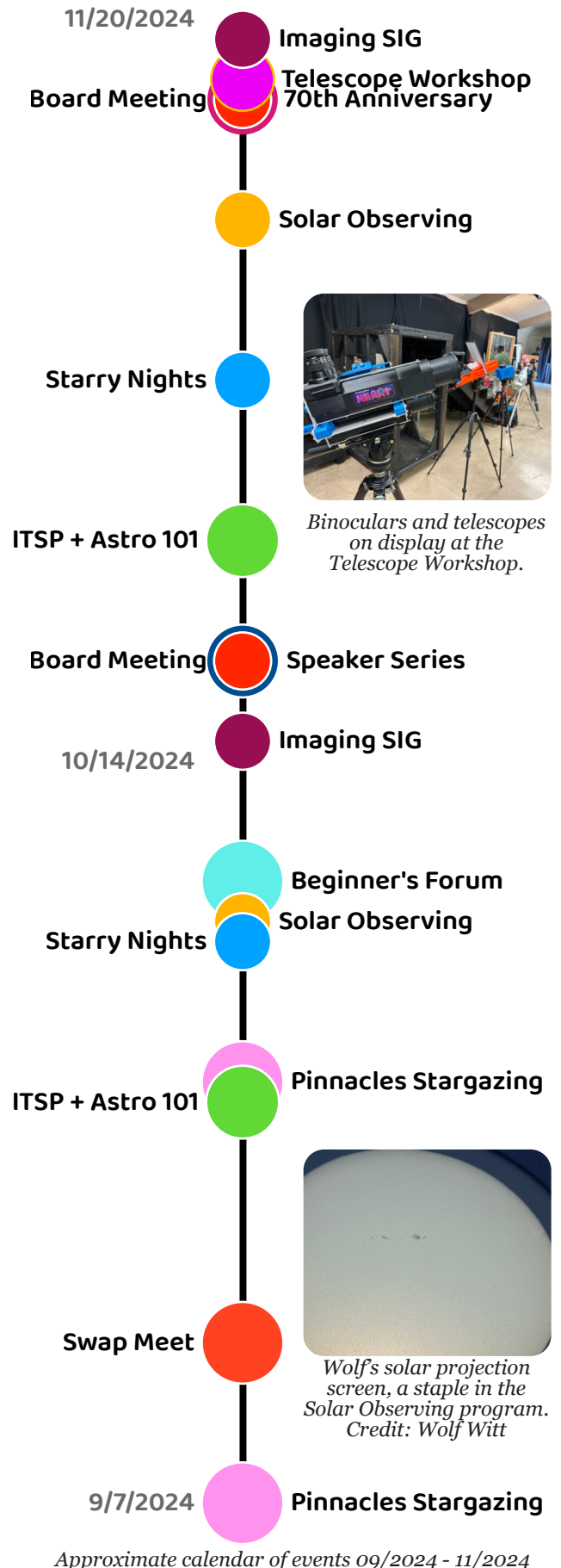
Starry Nights is held every month, weather permitting, at Rancho Cañada Del Oro Open Space Preserve (RCDO). This public star party is a collaboration between the Open Space Authority (OSA) and SJAA. This past quarter, members of the public (usually 80-100 people per event) were treated to views of classic late-Summer and early-Fall celestial targets, including the Hercules Cluster, the Ring Nebula, and the Dumbbell Nebula. We got our first views of Andromeda for the season, along with the welcome return of Saturn and Jupiter to the evening skies. Sign up to attend any future session on our Meetup page!

In-Town Star Party and Astro 101

On one Friday each month, we host a star party right from San Jose at our home base of Houge Park. Ten to fifteen volunteers help show crowds upwards of 100 people the sights of the cosmos. ITSPs this quarter were preceded by Astronomy 101 talks, "Sights of the Cosmos" and "Intro to Observing." These talks were given by Wolf and Carl S. to appreciative members of the public. Look for ITSP and Astro 101 events on Meetup!

Beginner's Astronomy Forum

This past quarter, we hosted another Beginner's Astronomy Forum, which is usually held about every two months. These sessions are held online, over Zoom, and this time around we had ten or so participants asking whatever questions come to mind. The forum is designed to be open-ended and give new star gazers a safe space to ask questions and get answers!



Ready Now

Results from Our Satisfaction Surveys!

Marianne Damon

We use the facilities at Houge Park thanks to the City of San Jose's Parks, Recreation, and Neighborhood Services (PRNS). In return, there are some requirements that we are contractually obligated to fulfill. We collect participant sign-in sheets and satisfaction surveys for our activities hosted at Houge, and we let PRNS know of all our upcoming activities at least a month in advance. Likewise, we report attendance information and hours of direct and indirect activity each month for the past month's events. "Direct hours" is time spent delivering programs to the public. "Indirect hours" is time meeting and preparing for activities for both public events and member-only events at Houge. Every six months we prepare and submit a status report.

On October 15 we posted our report for April 1 through September 30, 2024. We held 21 public events: 6 Astronomy 101, 6 ITSP, 5 Solar, 3 Speaker, and 1 Special Event/Swap Meet. This represents 35 direct hours. We have three General Goals, set by PRNS for contracted programs at all parks. The first: 85% of participants will rate services as good or excellent. We had 97 surveys; 5 good and 92 excellent, for a result of 100%. The second: 85% of participants will rate the facilities as clean and well-maintained. We had only yes responses. The third: 75% of participants will come from San Jose zip codes. Our result was 57%; we explained that we do have good representation of participants from surrounding zip codes, but we have many people who come from all over Santa Clara County and beyond. Per our sign-in sheets and zip code analysis, we had 485 unduplicated participants (we subtract out individuals who come to more than one event, a PRNS reporting requirement!). We know that not everyone signs in, so our total is, in actuality, higher than this.

For our Program Specific Goals, for each program we have set the goal that 85% of the participants will rate the event as good or excellent. All our programs met the goal at 100%. To the right are the programs, their ratings, and some comments.

Who does all this work? First, thanks to Program Leads and their volunteers who handle the sign-in sheets and satisfaction surveys! Marianne Damon provides the copies to the programs, collects them, tallies the sign-in sheets and sends them to Renuka, tallies the satisfaction surveys, and figures the unduplicated counts. Renuka Veeraswami prepares the upcoming calendar for PRNS, enters the direct and indirect hours each month, the total attendance, and the unduplicated attendance. She also enters each person's zip code in a spreadsheet. Rob Jaworski, Renuka and Marianne prepare the reports and interact with PRNS, which includes a once-a-year site visit.

Our BIG GOAL for this year is to automate the entry for

sign-in sheets and satisfaction surveys. We hope to establish QR codes for gathering this information by participants using their phones. By feeding into a database, we hope to cut down on the time and effort required to manage the data. We are so pleased that Radhika Gupta has volunteered to help with this endeavor. We welcome anyone else who would like to help!

SPEAKER NIGHT

13 surveys, 2 good, 11 excellent

- The speaker was very knowledgeable and the talk was held in a great venue.
- Thank you for such inspiring talk (and snacks!).
- High quality talk, good A-V.

IN TOWN STAR PARTY (ITSP)

26 surveys, 1 good, 25 excellent

- Extremely fun; learning about objects in the sky has been fun and interesting!
- It's so amazing to see City doing this program and helping residents to explore astronomy. A great job by team here!
- Thank you for letting us see the stars and explaining. Very friendly team.

SOLAR OBSERVING

25 surveys, 1 good, 24 excellent

- Very interesting talk. Great for kids and adults!
- Good explanations, friendly docents, good equipment.
- Highly informative. Lots of friendly and enthusiastic guidance.

ASTRONOMY 101

33 surveys, 1 good and 32 excellent

- Super educational; highly recommend to any audience.
- Excellent communication of a lot of information using verbal and visual interest.
- Informative, entertaining, and thought-provoking, Loved it!

School Star Party Award

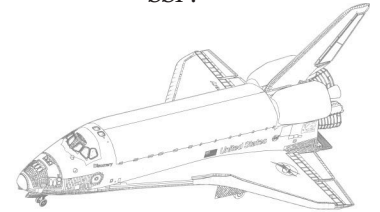
This summer, Jon Anderson accepted an award from the PTA of Fremont for his and SJAA's work with Fremont schools via the School Star Party Program (SSP). Recipients received a living plant as thanks for helping to grow their community. As Jon puts it, "This award serves to show the great interest in our program & the gratitude of the schools, students & teachers." Congrats and thanks to Jon and all the volunteers who make SSP possible!



Jon Anderson (middle) poses with his living plant; a token of appreciation from the Fremont schools.



The certificate awarded to Jon and the SSP.



Swap Meet Recap

Rob Jaworski

Seventy years is a long stretch of time for a small, local astronomy club, so let me take this opportunity to say Happy Birthday to the SJAA! I'm glad to be a part of it, and my tenure of over ten years has made me sort of an historical sink. I don't know a lot of what the club was like in the days before the early aughts, but I do have some familiarity with how things have been since I started.

Two of those things are services that the organization provides for its members as well as the local amateur astronomy community, things which brought both a social and practical value. For years, the SJAA conducted an annual auction. This was a lively affair complete with an auctioneer, bidding and the whole nine yards. However, the administrative effort for this event became too great and so the club's leaders decided to suspend it indefinitely. In its place, the club doubled the number of the other service, which is the annual swap meet. It used to be that the auction was held in the spring, and the swap meet in the fall (or vice versa), but when the auction was suspended, the swap meet became a twice annual event. Things were going great until the pandemic, which stopped all the club's activities.

In 2024, we began to see an increased level of interest for the swap meet. People would ask me and others when the

next swap will take place, we'd see such queries on local mail lists and everywhere else. So we decided to make it happen in the fall of this year. On 15 September 2024, we held our first swap meet since before the pandemic. People were really happy about that and it satisfied a built up demand since people had a lot of unused astronomy (and related!) gear gathering dust in their closet. The other thing it satisfied was a need for socialization, and let me say that this was probably the more important aspect of holding the swap. New friendships were made and old ones were rekindled as people who either never met or hadn't seen one another in years all came together to enjoy a pleasant Sunday afternoon 'talking shop'.

The event was well attended, perhaps a hundred people came through in the three hours that we opened the Hall at San Jose's Hogue Park. As the event wound down, the expected question was inevitably raised, When will the next swap meet be? Before the pandemic, the swap was biannual, and we see no reason for that to change. We plan to bring it back in all its glory, once in the spring and again the fall of each year. If you were not able to join us in September, keep an eye out in the Ephemeris or an SJAA email list where the exact date will be announced. We are looking forward to it and hope to see you there!

Spot the King of Planets!

Dave Prosper via Night Sky Network



NASA's Juno mission captured this look at the southern hemisphere of Jupiter on Feb. 17, 2020, during one of the spacecraft's close approaches to the giant planet. This high-resolution view is a composite of four images captured by the JunoCam imager and assembled by citizen scientist Kevin M. Gill.

JUPITER IS OUR solar system's undisputed king of the planets! Jupiter is bright and easy to spot from our vantage point on Earth, helped by its massive size and banded, reflective cloud tops. Jupiter even possesses moons the size of planets: Ganymede, its largest, is bigger than the planet Mercury. What's more, you can easily observe Jupiter and its moons with a modest instrument, just like Galileo did over 400 years ago.

Jupiter's position as our solar system's largest planet is truly earned; you could fit 11 Earths along Jupiter's diameter, and in case you were looking to fill up Jupiter with some Earth-size marbles, you would need over 1300 Earths to fill it up – and that would still not be quite enough! However, despite its formidable size, Jupiter's true rule over the outer solar system comes from its enormous mass. If you took all of the planets in our solar system and put them together, they would still only be half as massive as Jupiter all by itself. Jupiter's mighty mass has shaped the orbits of countless comets and asteroids. Its gravity can fling these tiny objects towards our inner solar system and also draw them into itself, as famously observed in 1994 when Comet Shoemaker-Levy 9, drawn towards Jupiter in previous orbits, smashed into the gas giant's atmosphere. Its multiple fragments slammed into Jupiter's cloud tops with such violence that the fireballs and dark impact spots were not only seen by NASA's orbiting Galileo probe but also by observers back on Earth!

Jupiter is easy to observe at night with our unaided eyes, as well-documented by the ancient astronomers who carefully recorded its slow movements from night to night. It can be one of the brightest objects in our nighttime skies, bested only by the Moon, Venus, and occasionally Mars, when the red planet is at opposition. That's impressive for a planet that, at its closest to Earth, is still over 365 million miles (587 million km) away. It's even more impressive that the giant world remains very bright



Look for Jupiter near the Eye of the Bull, Aldebaran, in the Taurus constellation on the evening of December 15, 2024. How many moons can you count? Credit: Stellarium Web

to Earthbound observers at its furthest distance: 600 million miles (968 million km)! While the King of Planets has a coterie of 95 known moons, only the four large moons that Galileo originally observed in 1610 – Io, Europa, Ganymede, and Calisto – can be easily observed by Earth-based observers with very modest equipment. These are called, appropriately enough, the Galilean moons. Most telescopes will show the moons as faint star-like objects neatly lined up close to bright Jupiter. Most binoculars will show at least one or two moons orbiting the planet. Small telescopes will show all four of the Galilean moons if they are all visible, but sometimes they can pass behind or in front of Jupiter or even each other. Telescopes will also show details like Jupiter's cloud bands and, if powerful enough, large storms like its famous Great Red Spot, and the shadows of the Galilean moons passing between the Sun and Jupiter. Sketching the positions of Jupiter's moons during the course of an evening – and night to night – can be a rewarding project! You can download an activity guide from the Astronomical Society of the Pacific at bit.ly/drawjupitermoons.

Now in its eighth year, NASA's Juno mission is one of just nine spacecraft to have visited this impressive world. Juno entered Jupiter's orbit in 2016 to begin its initial mission to study this giant world's mysterious interior. The years have proven Juno's mission a success, with data from the probe revolutionizing our understanding of this gassy world's guts. Juno's mission has since been extended to include the study of its large moons, and since 2021 the plucky probe, increasingly battered by Jupiter's powerful radiation belts, has made close flybys of the icy moons Ganymede and Europa, along with volcanic Io. What else will we potentially learn in 2030 with the Europa Clipper mission?

Find the latest discoveries from Juno and NASA's missions to Jupiter at science.nasa.gov/jupiter/

C/2023 A3

Tsuchinshan-ATLAS

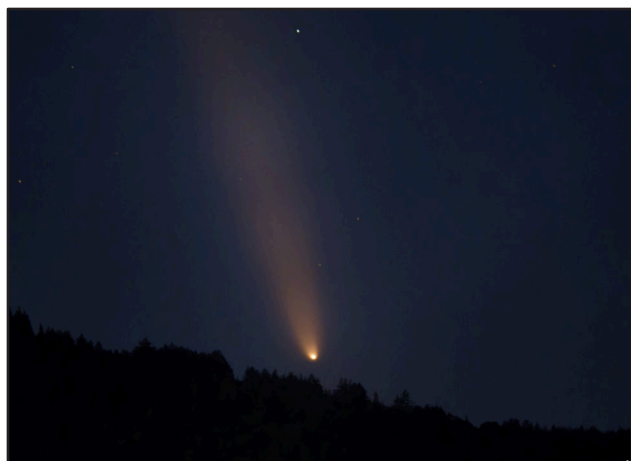
This quarter, several of our members captured wonderful images of comet C/2023 A3 Tsuchinshan-ATLAS. From Francesco's cover photo, with its detailed tail against a tapestry of picture-perfect stars, to Yu Li's run-in with a Starlink train (right) and Rajah's early June sighting, these images show the comet throughout the last six months. Carl's images (bottom) show the comet as visible from the Sierras a few weeks after perihelion (the comet's closest approach to the sun), which was on September 27th. At that time the comet was just 0.39 AU, closer to the sun than the Earth. As the comet slowly appeared in the evening sky after perihelion, it was visible to the naked eye as dusk turned to dark in the South-Southwest.

C/2023 A3 is arguably in the "great comet" class - comparable to the 2020 NEOWISE comet, and Hale-Bopp in 1997. Great comets are loosely defined as "bright enough to be noticed by casual observers who are not looking for them" [Wikipedia]. Tsuchinshan-ATLAS certainly falls into this category by that definition. Observers far and wide noted the bright comet in the waning daylight hours shortly after perihelion with its long tail visible to the naked eye.

However you slice it, this comet was a delight!



Yu Li's image taken from Lake San Antonio on Oct 19th using a Canon R6 with Canon EF 70-300mm f/4-5.6 L lens on an iOptron CEM40 mount. One exposure at 30s and 1600 ISO.



Left: Ed Wong captured the comet from Gilroy on October 13th at 7:57pm, just as the comet started sinking below the horizon. This image is a single, two-second exposure.

Right: this capture of C/2023 A3 back in June by our very own Rajah Chandrasekhar from RCDO shows the comet before reaching full brightness.



Two images of the comet taken from the Sierras by Carl Svensson on October 14th.

Left: a single, 30-second exposure taken with a Nikon D7000 and Nikkor lens.

Right: about 11 minutes of 30-second sub exposures taken with an 80mm refractor and ASI1600MM camera with RGB filters.



